Experiment 3

Flow and Thermal Behaviour in a Centrifugal Pump Passage

Aim

To evaluate centrifugal pressure development and temperature variation in a simplified rotating annular passage at several pump speeds.

Simulation Parameters

Parameter	Value
Coordinate system	Cylindrical
Radial grid points	80
Azimuthal grid points	160
Reynolds number	500
Iterations	1200
Pump speeds	1000, 2000, 3000 rpm

Numerical Method

A reduced-order model in a rotating polar frame prescribes radial and tangential velocity components. The model includes centrifugal pressure rise, wall heat exchange, and heating associated with losses.

Boundary Conditions

The inlet has a specified temperature and radial inflow at the pump eye. Flow moves outward radially, the azimuthal direction is periodic, and wall heat transfer is represented by a distributed source term.

Observations

Pressure contours, estimated pump head, absolute and relative speed, inlet/outlet temperatures, and thermal residual histories were compared across the three rotational speeds.

Results

Speed (rpm)	Re	Inlet temp. (K)	Outlet temp. (K)	Thermal residual	Peak pressure scale
1000	500	300	300.30	3.56 × 10■■	7 × 10■
2000	500	300	300.18	1.81 × 10■■	2.8 × 10■
3000	500	300	300.12	1.22 × 10■■	2.8 × 10■

Conclusion

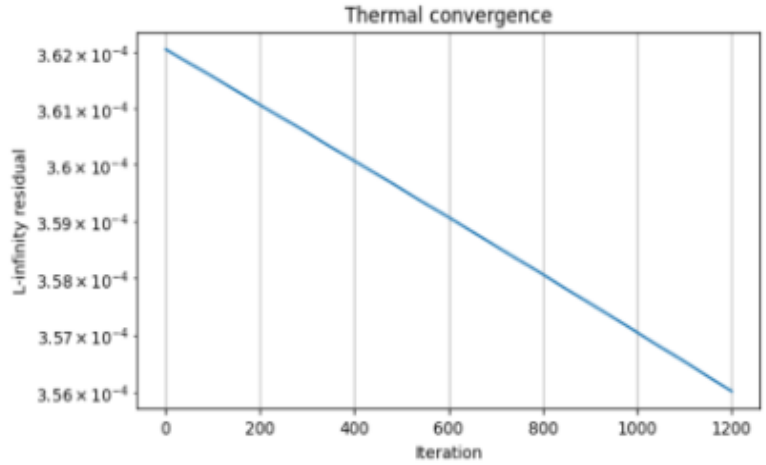
The pressure field strengthens as rotational speed increases, consistent with greater centrifugal pressure generation. The reported maximum pressure scale rises from about 7 × 10■ at 1000 rpm to about 2.8 × 10■ at 2000 and 3000 rpm. The predicted outlet-temperature increase remains modest, and the thermal residual decreases across the tested speeds.

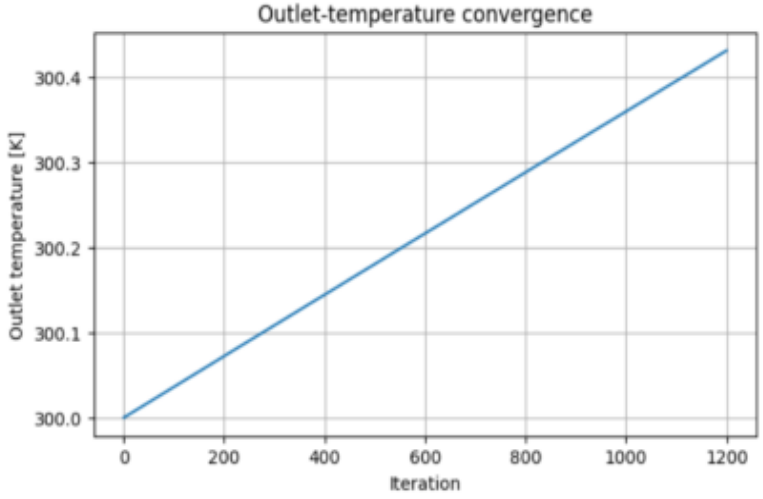
Original Simulation Figures

The following pages contain the plots and simulation outputs from the supplied report.

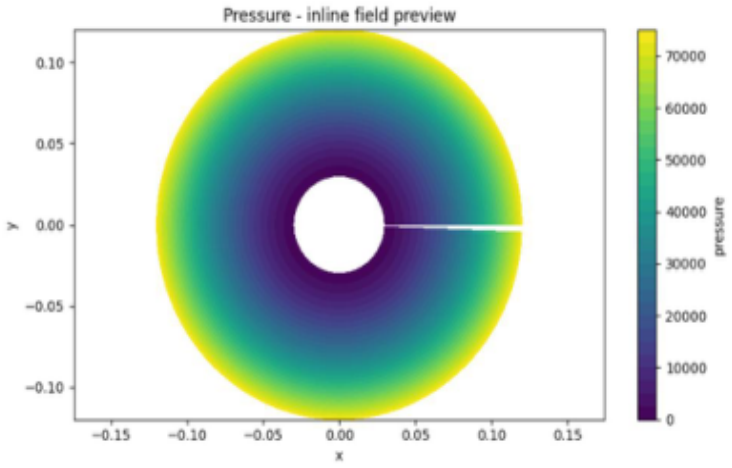
Original simulation output (source page 2)

For pump rotation at 1000 rpm:

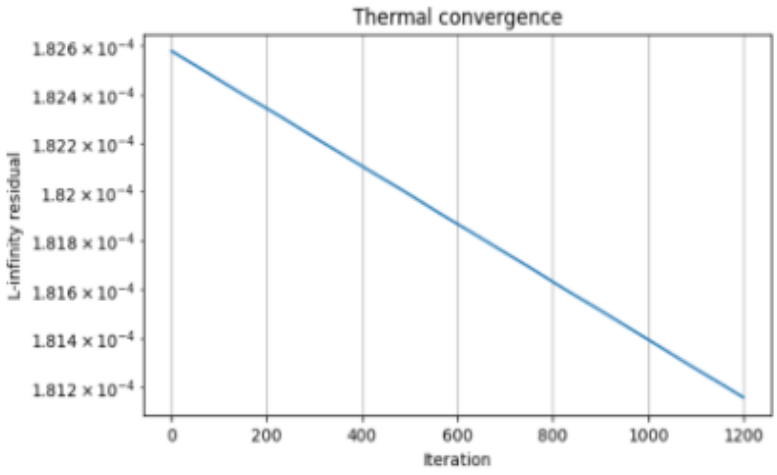




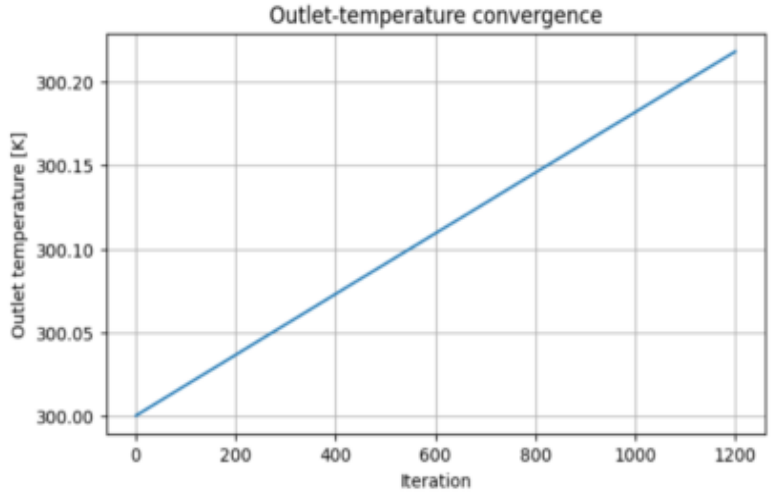
Original simulation output (source page 3)

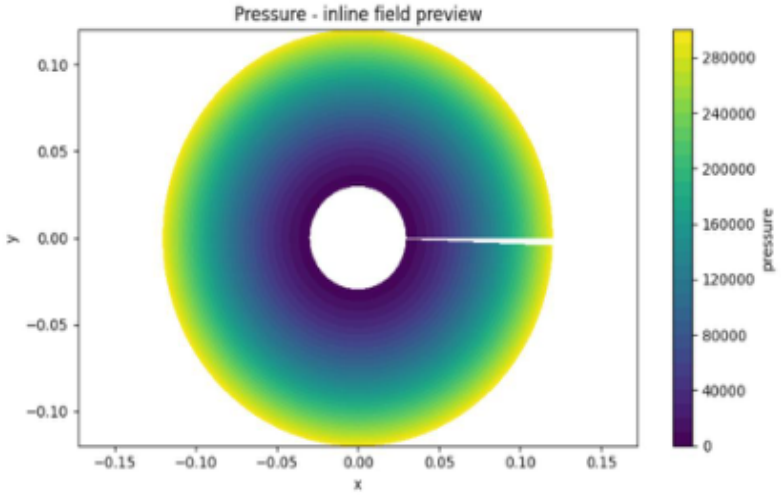


For pump rotation at 2000 rpm:



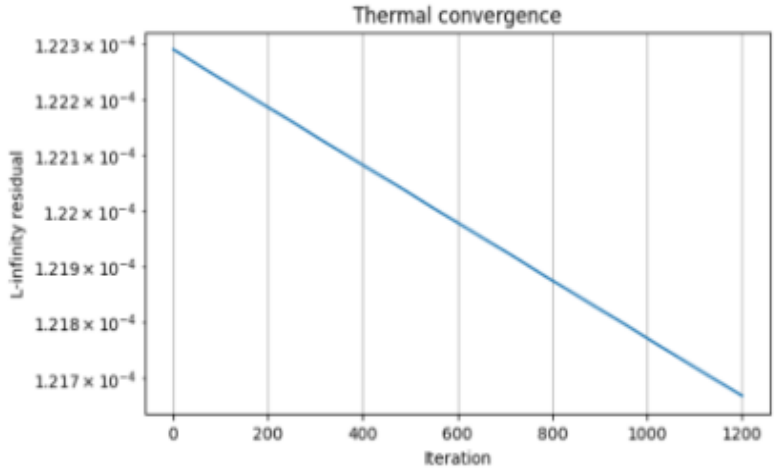
Original simulation output (source page 4)

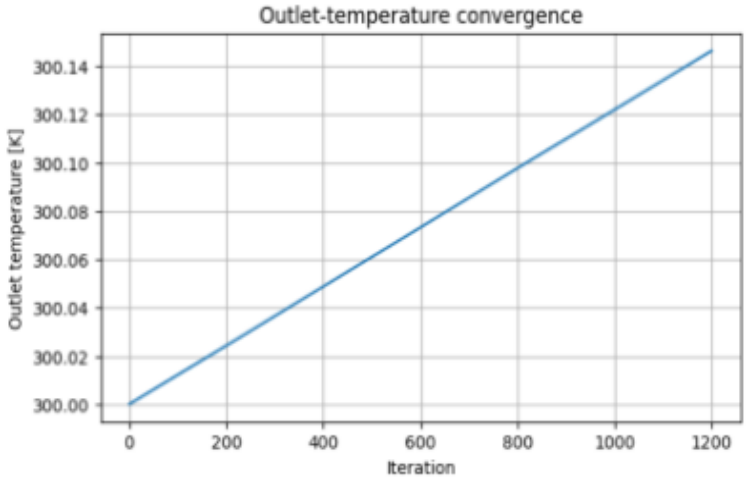




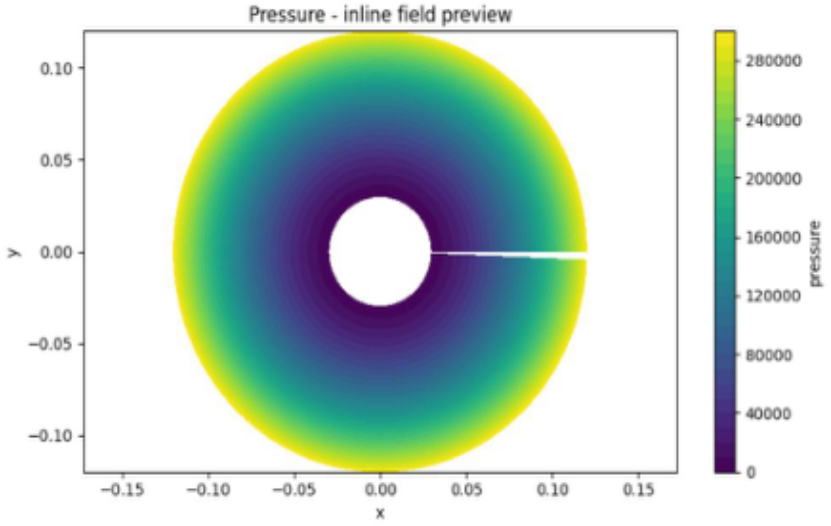
Original simulation output (source page 5)

For pump rotation at 3000 rpm:





Original simulation output (source page 6)



Result:

Pump Speed (RPM)	Reynold's number	Inlet temperature (k)	Outlet temperature (k)	Thermal residual	Max pressure from contour
1000	500	300	300.3	$3.56x10^{-4}$	$7x10^{4}$
2000	500	300	300.18	$1.81x10^{-4}$	$2.8x10^{5}$
3000	500	300	300.12	$1.22x10^{-4}$	$2.8x10^{5}$

Conclusion:

The centrifugal pump simulations were performed at 1000, 2000, and 3000 rpm while maintaining a constant Reynolds number of 500. The inlet temperature was maintained at approximately 300 K for all cases. At 1000 rpm, the outlet temperature increases to approximately 300.30 K, corresponding to a temperature rise of approximately 0.30 K. Increasing the rotational speed to 2000 rpm results in an outlet temperature of approximately 300.18 K, while at 3000 rpm, the outlet temperature approaches approximately 300.12 K. The corresponding thermal residuals decrease from approximately 3.56×10^{-4} at 1000 rpm to 1.81×10^{-4} at 2000 rpm and 1.22×10^{-4} at 3000 rpm. The pressure field shows a substantial increase in centrifugal pressure rise with rotational speed, with the pressure scale increasing from approximately 7×10^{4} at 1000 rpm to approximately 2.8×10^{5} at the higher rotational-speed cases. These results demonstrate the strong influence of rotational speed on centrifugal pressure generation, while the predicted temperature variation remains relatively small, with the maximum observed temperature increase being approximately 0.30 K.